Bellabeat Smart Device Usage Analysis Case Study

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INTRODUCTION

Bellabeat is a high-tech company that manufactures health-focused smart devices designed to support women's wellness. The company collects data related to activity, sleep, stress, and other health metrics through its smart products and mobile application. By analyzing smart device usage data, Bellabeat aims to better understand consumer behavior and identify opportunities for business growth.

This case study analyzes publicly available smart device fitness data to identify trends in user behavior. The insights generated from this analysis will help guide Bellabeat's marketing strategy and support data-driven decision-making.

EXECUTIVE SUMMARY

This case study analyzes smart device usage data from Fitbit fitness trackers to identify patterns in user activity, sleep behavior, and sedentary time. The analysis combines Excel data cleaning, SQL querying, and Tableau visualization to explore trends across 1,373 daily activity records from approximately 35 users.

The results show that many users do not consistently reach the recommended daily activity level of 10,000 steps, with sedentary behavior accounting for the largest portion of recorded activity days. A strong positive relationship between daily steps and calories burned was observed, indicating that increased movement significantly contributes to higher energy expenditure. Sleep duration across users averaged between six and seven hours per night, remaining relatively consistent across activity levels.

Based on these findings, Bellabeat can improve user engagement by promoting increased physical activity, encouraging consistent sleep tracking, and providing personalized wellness insights through its smart device ecosystem. These strategies can help users build healthier habits while supporting Bellabeat's marketing and product development efforts.

ASK

Business Task

The objective of this analysis is to examine smart device usage data to identify trends in how consumers use fitness tracking devices. These insights will help Bellabeat understand user habits and inform marketing strategies for Bellabeat products.

Key Stakeholders

- Urška Sršen — Bellabeat cofounder and Chief Creative Officer
- Sando Mur — Bellabeat cofounder
- Bellabeat Marketing Analytics Team

Business Questions

1. What are some trends in smart device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat's marketing strategy?

PREPARE

Data Source

- The dataset used for this analysis is the **FitBit Fitness Tracker dataset** available on Kaggle.
  It contains personal fitness tracker data including **daily activity, steps taken, distance traveled, calories burned, and sleep patterns** collected from Fitbit devices.
- After merging the March–April and April–May activity datasets, the combined dataset contained **1373 daily activity records from 35 unique Fitbit users**. The original dataset documentation references 33 users; however, additional user identifiers appear in the second dataset, increasing the total number of unique users.
- Two datasets covering the periods **March–April 2016** and **April–May 2016** were downloaded and merged to create a single dataset for analysis.

Data Organization

The dataset is organized into multiple **CSV files**, each containing specific types of activity data such as daily activity, hourly activity, and sleep records.

Each record includes a **unique user identifier (Id)** along with timestamps and activity metrics such as steps, distance, calories burned, and active minutes. These files provide insight into how users interact with their fitness tracking devices throughout the day.

Data Credibility

The dataset is publicly available on Kaggle and contains **real smart device usage data collected from Fitbit users**.

The dataset is widely used for educational purposes in data analytics projects and allows analysts to explore patterns in physical activity and health-related behaviors.

However, because the dataset was collected from a limited group of users and several years ago, the results should be interpreted cautiously.

Data Limitations

Although the dataset provides valuable insights, it contains several limitations:

- The dataset includes activity data from only **35 users**, which represents a relatively small sample size.
- The data was collected in **2016**, which may not fully reflect current smart device usage trends.
- The dataset contains **limited demographic information**, making it difficult to analyze behavior by factors such as age, gender, or location.
- Results derived from this dataset may **not fully represent global smart device user behavior**.

PROCESS

The datasets were cleaned and prepared for analysis to ensure accuracy and consistency.

The following steps were performed:

• Verified the structure and format of the datasets.

• Combined the March–April and April–May activity datasets into a single dataset.

• Removed duplicate records using the Id and ActivityDate fields.

• Checked for missing or null values.

• Ensured consistency in column names and data formats.

• Converted date fields into appropriate date formats.

• Standardized the user ID format to prevent formatting inconsistencies.

• Created an ActivityCategory column based on daily step counts.

• Created a composite key (Id + Date) to uniquely identify each user-day record.

• Merged sleep data with activity data using the composite key.

These steps ensured that the data was accurate, consistent, and properly structured before performing the analysis.

ANALYSE

The cleaned dataset was analyzed to identify patterns in smart device usage behavior. The analysis focused on several key metrics including daily steps, calories burned, activity intensity, and sleep duration.

Daily Activity Trends

The average number of daily steps among users in the dataset is approximately 7,247 steps. This value is below the commonly recommended goal of 10,000 daily steps for maintaining an active lifestyle. This suggests that many users in the dataset are not consistently meeting recommended physical activity levels.

Activity Intensity

Analysis of activity minutes shows that users spend the majority of their time in sedentary behavior. On average, users spend approximately 993 minutes per day sedentary, compared to only 19 minutes of very active activity and 13 minutes of fairly active activity. This indicates that high-intensity physical activity is relatively limited among users.

Activity Level Classification

To better understand user behavior, daily step counts were categorized into four activity levels based on thresholds proposed by Tudor-Locke & Bassett (2004): Sedentary, Low Active, Moderately Active, and Very Active. This classification helps identify patterns in user activity levels and highlights opportunities to promote healthier behaviors.

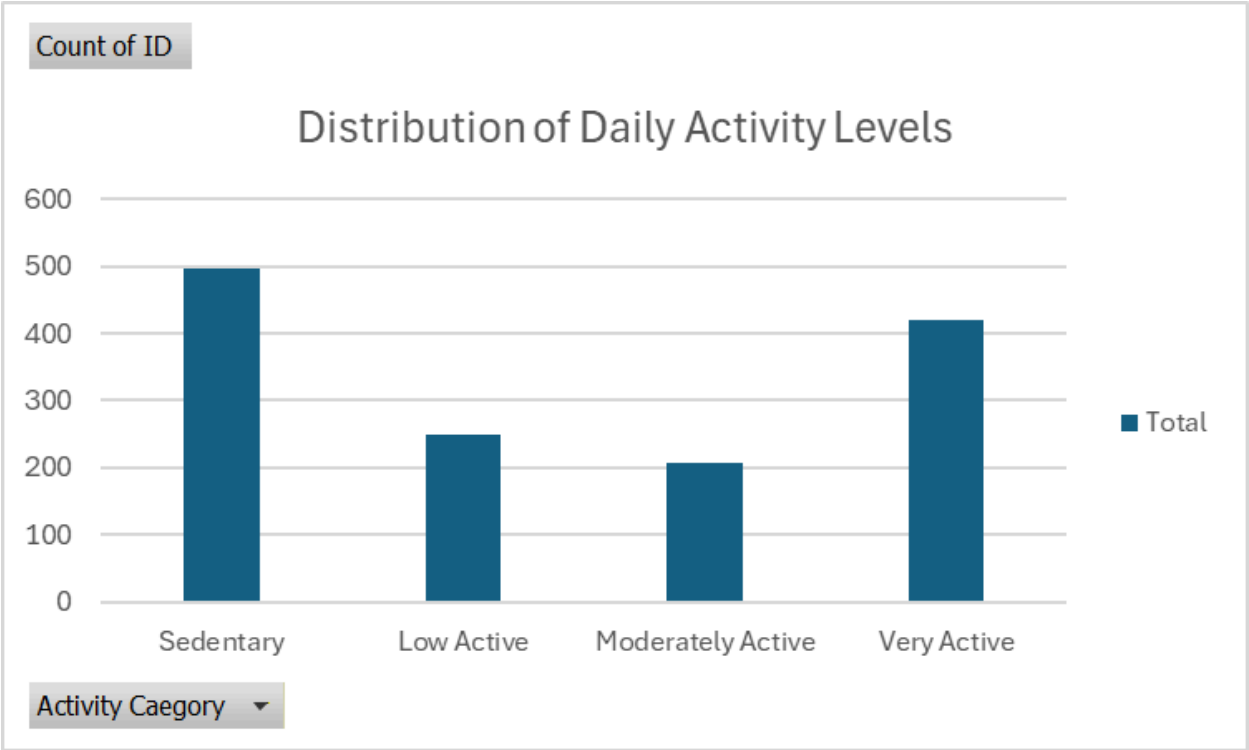
Relationship Between Steps and Calories

A scatter plot comparing daily steps and calories burned shows a clear positive relationship between physical activity and energy expenditure. As the number of daily steps increases, the number of calories burned also increases, indicating that increased movement contributes to higher caloric expenditure.

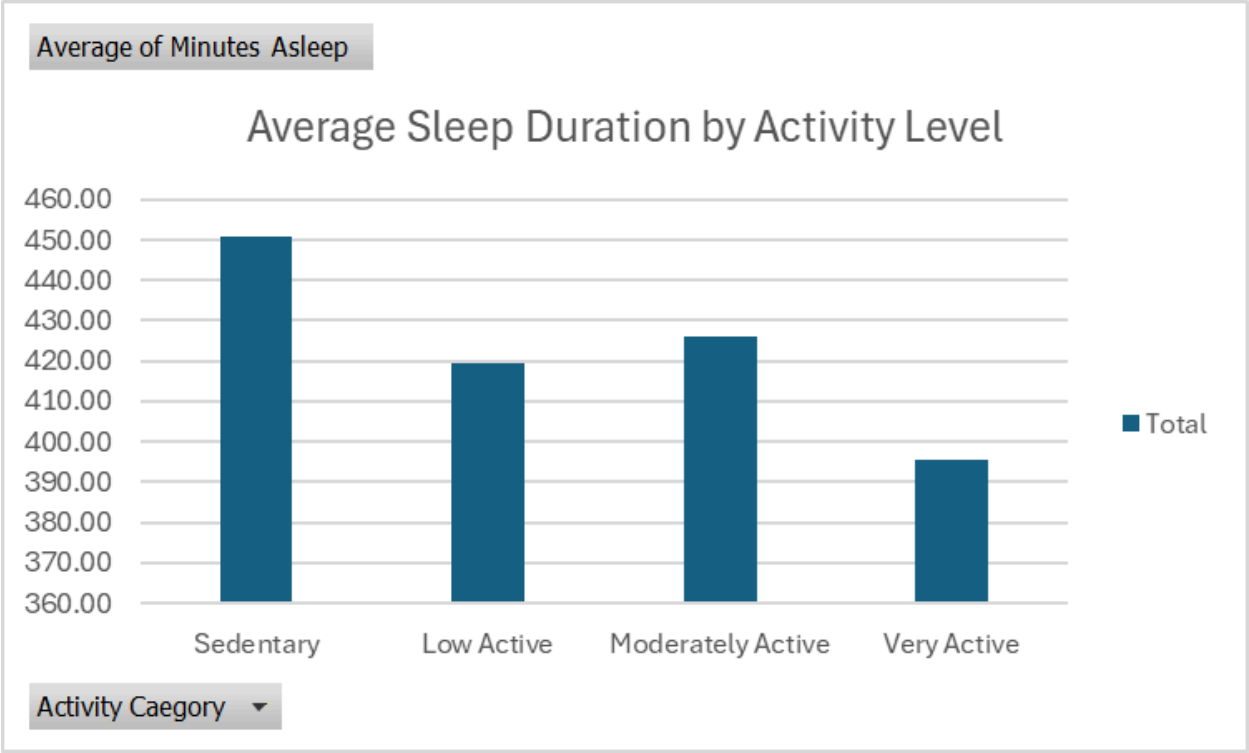
Sleep Patterns

Analysis of sleep data indicates that users sleep approximately **6–7 hours per night on average**. Users spend an average of **7.6 hours in bed**, suggesting that roughly **39 minutes are spent awake before falling asleep or during nighttime awakenings**. Overall, sleep duration among users appears relatively consistent, though it remains slightly below the recommended **7–9 hours of sleep per night** for adults.

A scatter plot comparing daily steps and calories burned shows a clear positive relationship between physical activity and energy expenditure.

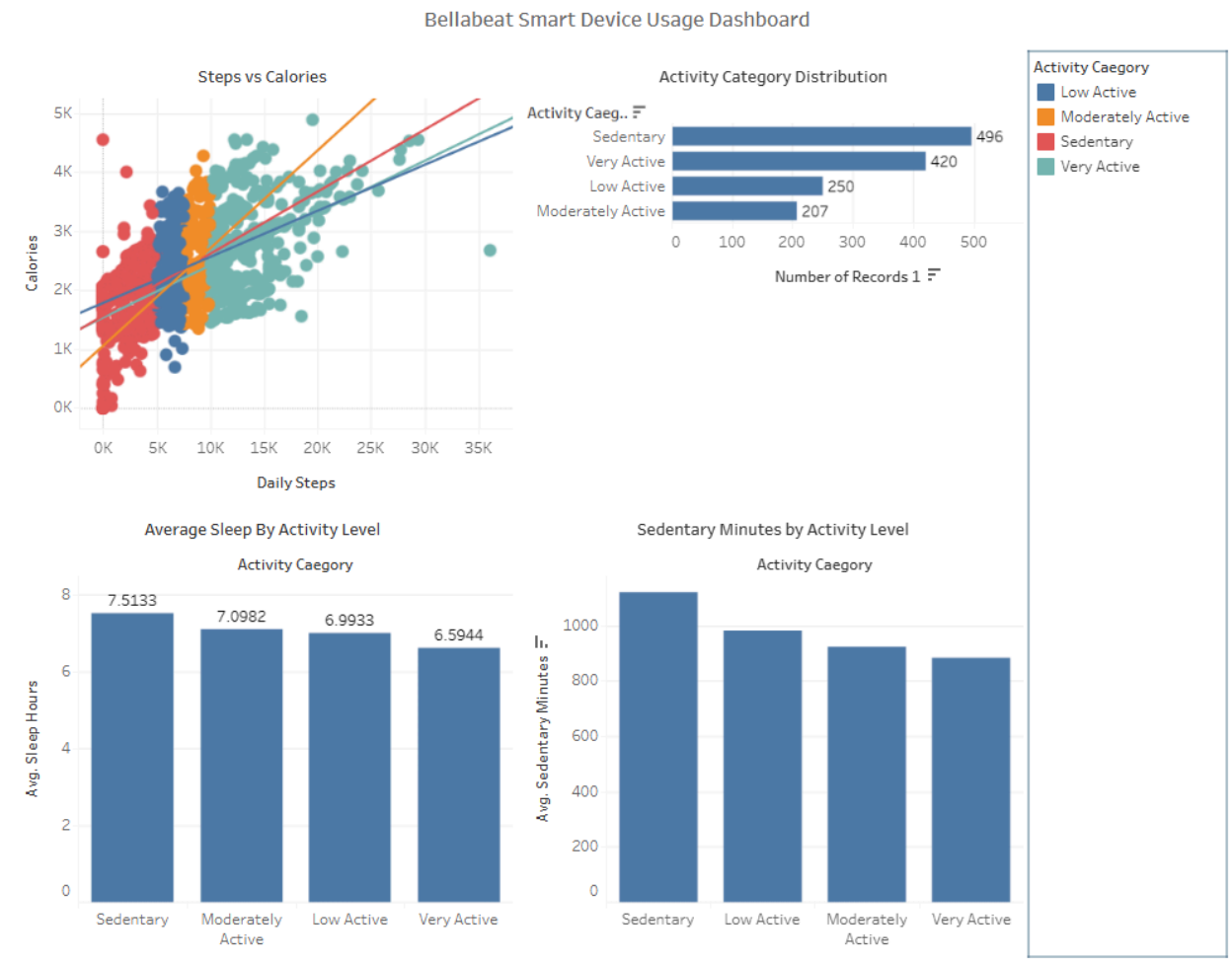


The distribution of activity levels shows that sedentary activity accounts for the largest share of recorded days.



The chart illustrates that sedentary users recorded slightly higher average sleep durations compared to more active users, although the difference across categories is relatively small.

The following visualizations were created in Tableau to support the analysis:



SHARE

The visualizations created in Tableau highlight several important patterns in user activity, sleep behavior, and sedentary time.

The distribution of activity levels shows that **sedentary activity accounts for the largest share of recorded days**, indicating that many users spend significant portions of their day inactive.

The analysis of sleep patterns suggests that **average sleep duration remains relatively consistent across different activity levels**, with most users sleeping between **6 and 7.5 hours per night**.

Key insights from the analysis include:

1. **Sedentary activity levels represent the largest share of recorded activity days.**

2. **Many users do not consistently reach the recommended 10,000 daily steps**, indicating opportunities to promote more active lifestyles.

3. **Higher activity levels are associated with increased calorie expenditure**, demonstrating the positive relationship between movement and energy use.

4. **Sleep duration remains relatively stable across activity categories**, suggesting that activity levels may not significantly impact sleep duration within this dataset.

5. **Sedentary time remains high even among more active users**, indicating potential opportunities to encourage more frequent movement throughout the day.

These insights provide a foundation for developing strategies that encourage healthier lifestyle habits among smart device users.

ACT

Based on the analysis, several recommendations can help Bellabeat improve **user engagement, encourage healthier habits, and strengthen its marketing strategy.**

Recommendation 1

Bellabeat could encourage users to increase daily physical activity through personalized step goals, reminders, and activity challenges within the app.

Recommendation 2

Bellabeat could promote sleep tracking features by providing bedtime reminders and sleep quality insights, helping users maintain healthy sleep habits.

Recommendation 3

Bellabeat could highlight the relationship between activity and overall wellness in its marketing campaigns to encourage users to track both physical activity and sleep behavior consistently.

Recommendation 4

Bellabeat could develop engagement features such as daily wellness summaries, encouraging users to maintain balanced routines that include movement, rest, and recovery.

KEY INSIGHT SUMMARY

- Many users do not reach the recommended 10,000 daily steps.
- Sedentary activity levels account for the largest share of activity days.
- Higher step counts are associated with increased calorie expenditure.
- Sleep duration is relatively consistent across activity levels.

CONCLUSION

This case study analyzed smart device usage trends using Fitbit fitness tracker data to better understand user activity patterns, sleep behavior, and sedentary habits. The analysis revealed that many users do not consistently meet recommended daily activity levels, with sedentary behavior representing the largest portion of recorded activity days. Additionally, while sleep duration remained relatively consistent across activity levels, it was slightly below recommended health guidelines for many users.

The analysis also demonstrated a clear relationship between **daily steps and calories burned**, reinforcing the importance of regular physical activity for overall health and wellness.

These insights provide valuable opportunities for Bellabeat to enhance its product experience and marketing strategy. By encouraging increased movement, promoting consistent sleep tracking, and offering personalized wellness insights, Bellabeat can help users develop healthier daily routines.

Through data-driven decision-making, Bellabeat can continue to empower users to better understand their health behaviors while strengthening its position as a leading company in the smart wellness technology market.